Supplementary Material B: Simulation Supplement for "Latent Spatial Regimes in Spatio-Temporal State-Space Models: Evidence on Local Price Leadership in Italian Fuel Markets"

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This Supplementary Material collects all the material of the simulation study of Section 4 of the manuscript.

B1 Aims of the simulation study

SC-STEM allows every block of the parameters to differ across regimes. The simulation study asks three questions about the estimation procedure (i.e., the fit of the whole grid of (K, ϕ) followed by the two-step selection rule):

(Q1) **Complete heterogeneity.** When the data are generated by SC-STEM with every block of parameters different across regimes, does the procedure recover the number of regimes K, the partition of the locations and the regime-specific parameters Ψ_k?

(Q2) **Complete homogeneity.** When the data are generated by the pooled STEM model, how often does the procedure declare more than one regime?

(Q3) **One common block.** When one block of parameters (e.g., the regression coefficients, the persistence of the latent process, its innovation variance, the spatial range, or the whole measurement error) is common to all regimes and the others differ, how much does the recovery of the regimes change?

What a common block does to the estimator Given the partition, SC-STEM fits each regime on its own data only, so the sampling distribution of $\hat{\Psi}_k$ depends on the data and the parameters of regime k alone, and not on whether a block of Ψ_k equals the corresponding block of Ψ_l. A block common to all regimes in the data-generating process therefore does not make the other parameters better estimated: its only effect is to remove one source of contrast between the regimes, which can only make the partition harder to recover. Question (Q3) is read accordingly, as how much each block contributes to the separation of the regimes. Whether a block differs across regimes in a given data set is assessed by the bootstrap tests of the between-regime differences (Section 3.3 of the manuscript).

B2 The data-generating process

Every scenario is generated by one generator. A scenario is a choice of the parameter sets Ψ_1, Ψ_2, Ψ_3 of the three regimes; a variant is a choice of the dependence between regimes (Section B3.3). The

true number of regimes is $K = 3$ in every scenario except the pooled one, which has a single regime.

B2.1 The model simulated

Let $k_i \in \{1, \dots, K\}$ be the regime of location i, $i = 1, \dots, d$, and $\mathcal{C}_k = \{i : k_i = k\}$. For $i \in \mathcal{C}_k$ and $t = 1, \dots, T$, the SC-STEM model of Section 2.1 of the manuscript with one latent process ($p = 1$) and loading matrix $A = \mathbf{1}_d$ reads

$$z_{it} = x_{it}^\top \beta_k + y_t^{(k)} + e_{it}, \tag{B1}$$

$$y_t^{(k)} = G_k \, y_{t-1}^{(k)} + \eta_t^{(k)}, \qquad \eta_t^{(k)} \sim N(0, \sigma_{\eta k}^2), \tag{B2}$$

$$e_{t,\mathcal{C}_k} \sim N\big(0, \ \sigma_{\epsilon k}^2 I + \sigma_{\omega k}^2 \, C_{\theta_k}\big), \qquad [C_\theta]_{ij} = e^{-\theta h_{ij}}, \tag{B3}$$

with e_t independent over t, and with the latent processes and the error fields of different regimes independent of one another. Each regime carries $\Psi_k = (\beta_k, \sigma_{\epsilon k}^2, \sigma_{\omega k}^2, \theta_k, G_k, \sigma_{\eta k}^2, m_{0k})$. Because the regimes are independent, the log-likelihood of a partition is the sum of the log-likelihoods of the regimes, each the STEM log-likelihood of the locations of that regime alone. With $K = 1$ the model is the pooled STEM model.

Two sources of separation. Under Equations (B1)–(B3) the regimes differ in two ways: in their parameters Ψ_k, and because each has its own latent path $y^{(k)}$ and its own error field, independent of those of the other regimes. The second is a property of the model, not of the parameters: conditionally on the covariates, $\mathrm{Cov}(z_{it}, z_{js}) = 0$ for $i \in \mathcal{C}_k$, $j \in \mathcal{C}_l$, $k \neq l$, so two locations in different regimes are uncorrelated however close they are. The scenarios vary the first source; the variants of Section B3.3 switch the second on and off.

B2.2 Locations: four geometries

The locations lie in the plane, and distances are Euclidean in its units. The regime sizes are equal up to the remainder (balanced) or proportional to 1:2:3 (unbalanced), with no regime below six locations. The coordinates of a location of regime k are drawn as

$$s_i \mid k_i = k \ \sim \ N_2\big(\mu_k, \ \nu \, I_2\big), \qquad \mu_k = \frac{2\omega}{\sqrt{3}} \big(\cos a_k, \ \sin a_k\big), \quad a_k = 90^\circ, 210^\circ, 330^\circ, \tag{B4}$$

so that the three centers lie at the vertices of an equilateral triangle of side 2ω. The variance of the centers around their mean, per coordinate, is $c(\omega) = 2\omega^2/3$. Two rules fix the within-regime variance ν:

- *network spread fixed*: $\nu(\omega) = \nu_{\mathrm{tot}} - c(\omega)$, with $\nu_{\mathrm{tot}} = 0.4 + 2/3$, so that the variance of a coordinate over the whole network is ν_{tot} whatever ω is, and the regimes shrink as they move apart;
- *regime spread fixed*: $\nu = \nu(0.7) \approx 0.74$, so that the regimes keep the spread they have at $\omega = 0.7$ and the network widens instead.

The two rules coincide at $\omega = 0.7$. The design uses four geometries (Table B1, Figure B1). The degree of overlap is measured by the share of the locations of a regime that lie nearer the center of another regime than to their own; it is $2/3$ when the centers coincide and decreases as they move apart.

Total overlap, $\omega = 0$: the three regimes occupy the same area, and the partition has no spatial signature. The regimes can be told apart by their parameters and their independent dynamics only.

Medium-high overlap, $\omega = 0.7$: the regimes are contiguous areas that interpenetrate along their borders. This is the reference geometry.

Medium-low overlap, $\omega = 1$ with the regime spread fixed: the centers move apart while every regime keeps its spread, and the network covers a larger area.

Strong separation, $\omega = 1$ with the network spread fixed: the centers move apart and every regime becomes more compact.

In the pooled scenario there are no regimes, and the locations form one cloud, $s_i \sim N_2(0, \nu_{\text{tot}} I_2)$, covering the same area as the reference geometry: the network is not spatially clustered, so the selection rule receives no geometric cue.

Table B1: The four geometries of the regimes. Center distance: 2ω; regime sd: $\sqrt{\nu}$; distance in sd: $2\omega/\sqrt{\nu}$; nearer another center: share of the locations of a regime nearer the center of another regime; network sd: the standard deviation of a coordinate over the whole network.

geometry	ω	fixed spread	center distance	regime sd	distance in sd	nearer another center	network sd
total overlap	0.0	network	0.00	1.03	0.00	67%	1.03
medium-high overlap	0.7	network	1.40	0.86	1.63	32%	1.03
medium-low overlap	1.0	regime	2.00	0.86	2.32	20%	1.19
strong separation	1.0	network	2.00	0.63	3.16	10%	1.03

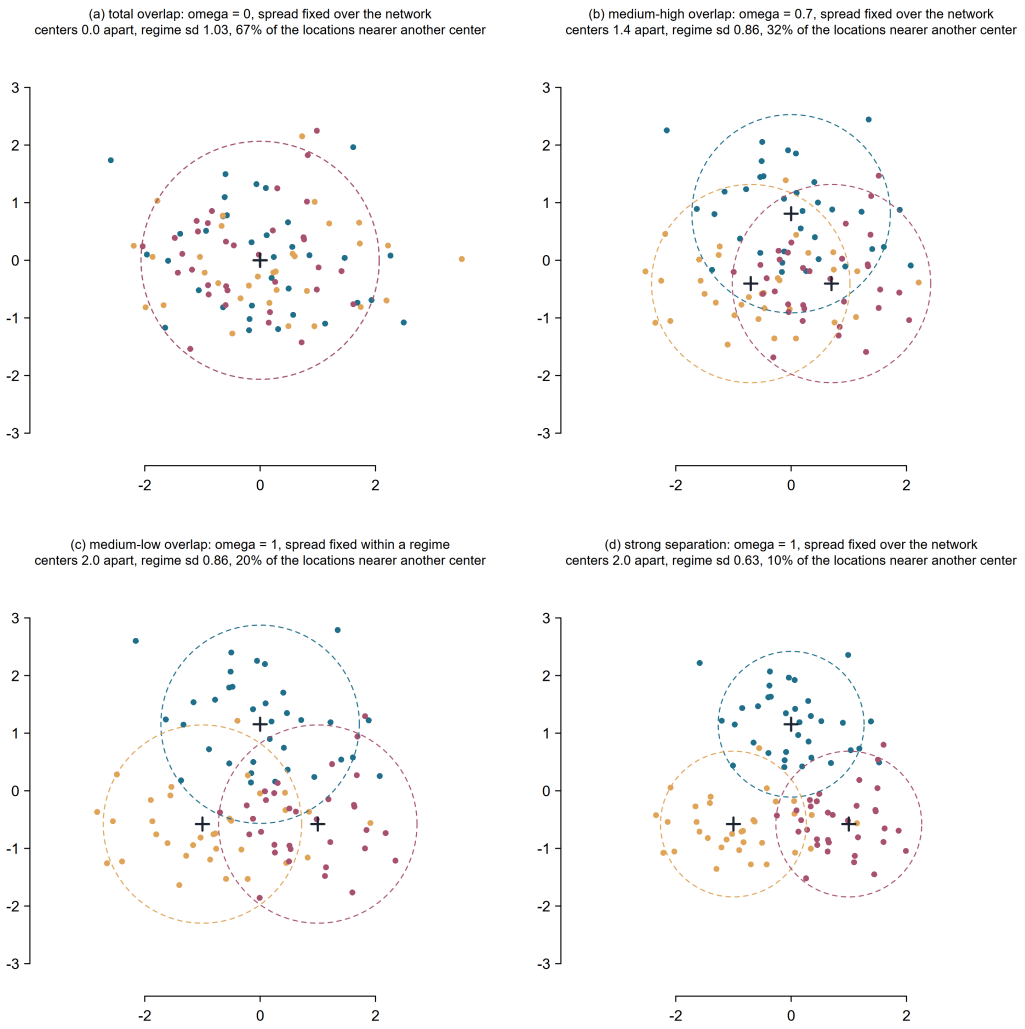


Figure B1: The four geometries, with $d = 100$ locations drawn as in the first replication of the study. Colors: regimes 1, 2 and 3; crosses: the centers μ_k; dashed circles: two within-regime standard deviations around each center.

B2.3 Covariate

One exogenous covariate, an AR(1) in time whose innovations are a spatially correlated field,

$$x_t = a\, x_{t-1} + \sqrt{1 - a^2}\; u_t, \qquad u_t \sim N_d(0, C_x), \qquad [C_x]_{ij} = e^{-3h_{ij}/R_x}, \tag{B5}$$

with $a = 0.7$, so that a shock to the covariate halves in about two periods, and practical range $R_x = 4$, the extent of the network, so that the covariate varies on a large spatial scale. The covariate is then centered and scaled over all its dT values. Its spatial correlation sits between two degenerate cases: a covariate independent across locations would make a contrast in its coefficient trivially visible, and one common to all locations would be indistinguishable from the latent process. The covariate vector is $x_{it} = (1, x_{it})^\top$ and $\beta_k = (\beta_{0k}, \beta_{1k})^\top$.

B2.4 Latent processes

One AR(1) per regime,

$$y_t^{(k)} = G_k\, y_{t-1}^{(k)} + \eta_t^{(k)}, \qquad \eta_t^{(k)} = \sigma_{\eta k}\left(\sqrt{\rho}\,\xi_t + \sqrt{1-\rho}\,\zeta_{kt}\right), \tag{B6}$$

with ξ_t and ζ_{kt} independent standard normal variables, independent over t and k. Then $\mathrm{Var}(\eta_t^{(k)}) = \sigma_{\eta k}^2$ and $\mathrm{Corr}(\eta_t^{(k)}, \eta_t^{(l)}) = \rho$ for $k \neq l$: ξ_t is the shock common to all regimes and ζ_{kt} the one specific to regime k. The processes start from zero and run through a burn-in of 200 periods, which is discarded.

Stationary moments. Writing $y_t^{(k)} = \sum_{j\geq 0} G_k^j \eta_{t-j}^{(k)}$,

$$v_k := \mathrm{Var}\big(y_t^{(k)}\big) = \frac{\sigma_{\eta k}^2}{1-G_k^2}, \qquad \mathrm{Corr}\big(y_t^{(k)}, y_t^{(l)}\big) = \rho\,\kappa_{kl}, \qquad \kappa_{kl} = \frac{\sqrt{(1-G_k^2)(1-G_l^2)}}{1-G_kG_l}. \tag{B7}$$

Since $(1-G_kG_l)^2 - (1-G_k^2)(1-G_l^2) = (G_k-G_l)^2$, $\kappa_{kl} \leq 1$ with equality if and only if $G_k = G_l$: two autoregressions with different persistence are not perfectly correlated even when they are driven by the same shocks. The dynamics of a regime are described by the pair (G_k, v_k), the persistence and the amplitude of the latent signal, and $\sigma_{\eta k}^2 = v_k(1-G_k^2)$ follows.

B2.5 Measurement error

Two constructions, independent over t.

Fields by regime. Each regime draws its own field, independently of the others, $e_{t,\mathcal{C}_k} \sim N(0, \sigma_{\epsilon k}^2 I + \sigma_{\omega k}^2 C_{\theta_k})$. Two locations in different regimes have uncorrelated errors.

One field. One spatial field spans the whole network and is scaled by regime,

$$e_{it} = \sigma_{\epsilon k_i}\,\varepsilon_{it} + \sigma_{\omega k_i}\,w_t(s_i), \qquad \varepsilon_{it} \sim N(0,1) \text{ independent}, \qquad w_t \sim N_d(0, C_\theta), \tag{B8}$$

so that $\mathrm{Cov}(e_{it}, e_{jt}) = \sigma_{\omega k_i}\sigma_{\omega k_j}\, e^{-\theta h_{ij}}$ also across the border between two regimes. A single field has a single range, so this construction requires θ to be common to all regimes.

Within each regime the two constructions give the same law, $N(0, \sigma_{\epsilon k}^2 I + \sigma_{\omega k}^2 C_\theta)$; they differ only in the covariance between regimes.

B2.6 Response and parametrization

The response is $z_{it} = \beta_{0k_i} + \beta_{1k_i}\, x_{it} + y_t^{(k_i)} + e_{it}$, and its conditional mean given the latent path, $\mu_{it} = \beta_{0k_i} + \beta_{1k_i} x_{it} + y_t^{(k_i)}$, is the target against which the fitted signal is scored. Each regime is described by

$$\big(\beta_{0k},\ \beta_{1k},\ G_k,\ v_k,\ \sigma_{\epsilon k}^2,\ \sigma_{\omega k}^2,\ R_k\big), \qquad \sigma_{\eta k}^2 = v_k(1-G_k^2), \qquad \theta_k = 3/R_k,$$

where R_k is the practical range, the distance at which the correlation of the spatial component falls to $e^{-3} \approx 5\%$. The five blocks of parameters used in Scenario S3 are $\beta = (\beta_0, \beta_1)$, G, σ_η^2, θ, and the error $(\sigma_\epsilon^2, \sigma_\omega^2, \theta)$.

B3 Scenarios and variants

A scenario fixes the parameter values of the three regimes, and a variant the dependence between them (Section B2). The names of the scenario-variants read as follows:

S1, S2, S3. The three families of scenarios, one for each question of Section B1: S1 is complete heterogeneity, S2 complete homogeneity, S3 one common block.

s, w. In S1 only, the level of separation of the regimes: strong (S1s) or weak (S1w).

beta, G, Seta, theta, error. In S3 only, the block of parameters that the three regimes share: the regression coefficients, the persistence of the latent process, the variance of its innovations, the range of the spatial field, or the whole measurement error.

ind, shr. After the hyphen, the dependence between regimes: independent regimes, or regimes that share the shocks of their latent processes and, where the range is common, one spatial field (Section B3.3).

B3.1 The baseline regime

Regime 1 of every scenario, and the only regime of the pooled scenario, carries the baseline: $\beta_0 = 2$, $\beta_1 = 0.55$, $G = 0.8$, $v = 0.30$ (so $\sigma^2_\eta = 0.108$), $\sigma^2_\epsilon = \sigma^2_\omega = 0.20$ and $R = 2$ ($\theta = 1.5$). The variance of the response at a location of regime 1 is $\beta_1^2 + v + \sigma^2_\epsilon + \sigma^2_\omega = 0.30 + 0.30 + 0.20 + 0.20 \approx 1$, so every contrast below reads in standard deviations of the response. The covariate and the latent process explain 30% of the variance each, the spatially structured and the purely local error 20% each; a latent shock halves in $\log 0.5 / \log 0.8 = 3.1$ periods, and the practical range is half the extent of the network.

B3.2 The scenarios

Table B2 gives the values of the three regimes of every scenario. The innovation variance of the latent process follows from its persistence and its stationary variance, $\sigma^2_\eta = v(1 - G^2)$, and the total error variance is $s^2 = \sigma^2_\epsilon + \sigma^2_\omega$. Figures B2–B4 show what the values of Table B2 do to the data.

Table B2: Values of regimes 1 / 2 / 3 by scenarios-variants. Regime 1 is the baseline in every scenario; S2 has one regime. "As S1s": the values of S1-strong. v: stationary variance of the latent process; R: practical range of the spatial field, $\theta = 3/R$. Last column: the variants run.

scenario	what it is	β_0	β_1	G	v	σ^2_ϵ	σ^2_ω	R	variants
S2	no regimes	2	0.55	0.8	0.30	0.20	0.20	2	one
S1s	every block differs, strongly	2 / 2.3 / 2.6	0.55 / 0.85 / 1.15	0.8 / 0.5 / 0.2	0.30 / 0.40 / 0.50	0.20 / 0.35 / 0.40	0.20 / 0.15 / 0.20	2 / 1.5 / 1	`ind`, `shr`
S1w	every block differs, weakly	2 / 2.1 / 2.2	0.55 / 0.65 / 0.75	0.8 / 0.7 / 0.6	0.30 / 0.35 / 0.40	0.20 / 0.25 / 0.30	0.20 / 0.175 / 0.15	2 / 1.5 / 1	`shr`
S3beta	as S1s, common coefficients	2	0.55	as S1s	as S1s	as S1s	as S1s	as S1s	`shr`
S3G	as S1s, common persistence	as S1s	as S1s	0.8	0.30 / 0.40 / 0.50	as S1s	as S1s	as S1s	`shr`
S3Seta	as S1s, common $\sigma^2_\eta = 0.108$	as S1s	as S1s	0.8 / 0.5 / 0.2	0.30 / 0.14 / 0.11	as S1s	as S1s	as S1s	`shr`
S3theta	as S1s, common range	as S1s	as S1s	as S1s	as S1s	as S1s	as S1s	2	`shr`
S3error	as S1s, common error	as S1s	as S1s	as S1s	as S1s	0.20	0.20	2	`shr`

S2: complete homogeneity. The data come from the pooled STEM model at the baseline,

$$z_{it} = 2 + 0.55\, x_{it} + y_t + e_{it}, \qquad y_t = 0.8\, y_{t-1} + \eta_t, \;\; \sigma^2_\eta = 0.108, \qquad e_t \sim N\big(0,\; 0.20\, I + 0.20\, C_{1.5}\big), \tag{B9}$$

with one latent path and one spatial field over the whole network. The scenario measures how often the procedure declares regimes that do not exist, that is the rate of false positives, as a function of the size of the network and of the length of the series. In the generator it is the corner in which the three parameter sets coincide, $\rho = 1$ and the field is one; with $\rho = 0$ or fields by regime it would no longer be the pooled model, so S2 has one variant only.

S1: complete heterogeneity. Every block of parameters differs across the three regimes, and the scenario measures whether the procedure recovers the number of regimes, the partition and the regime-specific parameters when the regimes exist. Regime 1 is the baseline, regime 3 carries the full contrast and regime 2 sits between them, at two levels: S1-weak (S1w), with poorly separated regimes, and S1-strong (S1s), with well-separated ones. Along the regimes the level rises, the covariate matters more, shocks die out faster but the latent signal is larger, and the error variance grows; the ranges decrease from half the network to a quarter of it. In S1s the variance of the response in regime 3 is $1.15^2 + 0.50 + 0.60 = 2.42$, against 1 in regime 1.

S3: one common block. One block takes the baseline value in all regimes and the other blocks keep the values of S1s, so that the comparison with S1s isolates the contribution of that block to telling the regimes apart. In S3beta the coefficients are common; in S3G the persistence $G = 0.8$ is common and the amplitudes v_k stay those of S1s; in S3Seta the innovation variance $\sigma_\eta^2 = 0.108$ is common and the persistences G_k stay those of S1s, so that the amplitudes follow from (B7); in S3theta the range $R = 2$ is common; in S3error the whole error, $\sigma_\epsilon^2 = \sigma_\omega^2 = 0.20$ and $R = 2$, is common. Making every block common does not give the pooled model: three independent copies of one STEM model on three sub-networks still have three latent paths and three fields. The pooled model is common parameters, plus one latent path, plus one field; three independent copies, moreover, remain separable by their dependence alone (Proposition B1).

How the variance of the response splits. Figure B2 splits the variance of the response of every regime among its four sources. In the baseline regime the four sources weigh about the same; along the regimes of S1-strong the covariate and the latent process weigh more and the response becomes more variable, while in S1-weak the regimes differ little. In S3-beta the covariate contributes the same variance to every regime, and in S3-error the error does.

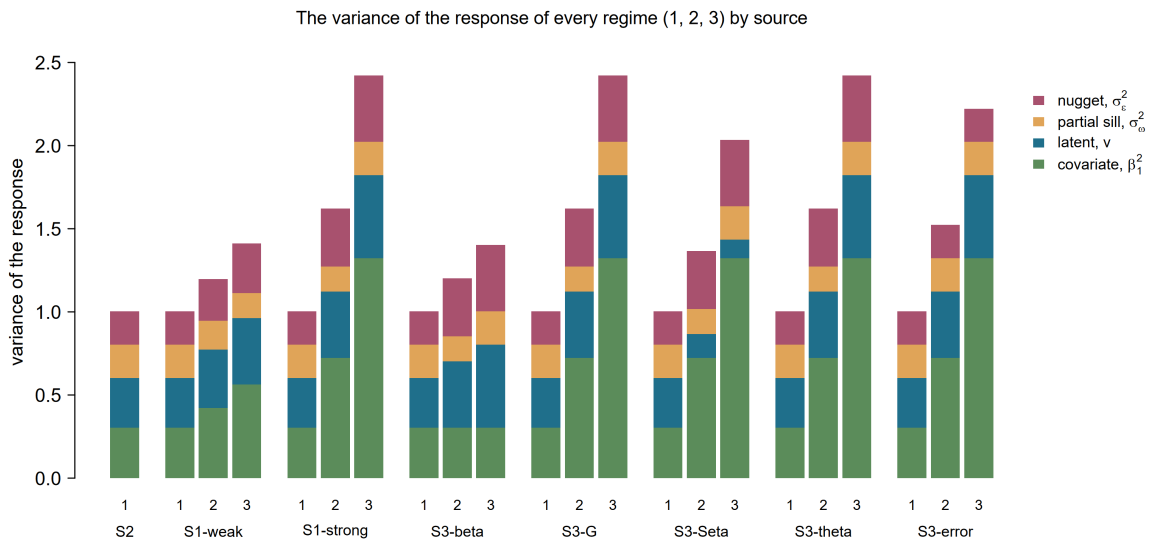


Figure B2: The variance of the response of every regime of every scenario, split into the variance due to the covariate (β_1^2, since the covariate is standardized), the latent process (v), the spatially structured error (the partial sill σ_ω^2) and the purely local error (the nugget σ_ϵ^2). The intercepts contribute no variance.

How the correlations decay. Figure B3 shows how the regimes differ in their dependence: the correlation of the error decays with distance faster in the regimes with a shorter range, and the autocorrelation of the latent process decays with the lag faster in the regimes with a smaller persistence. At the strong level the half-life of a latent shock falls from about three periods in regime 1 to less than one in regime 3.

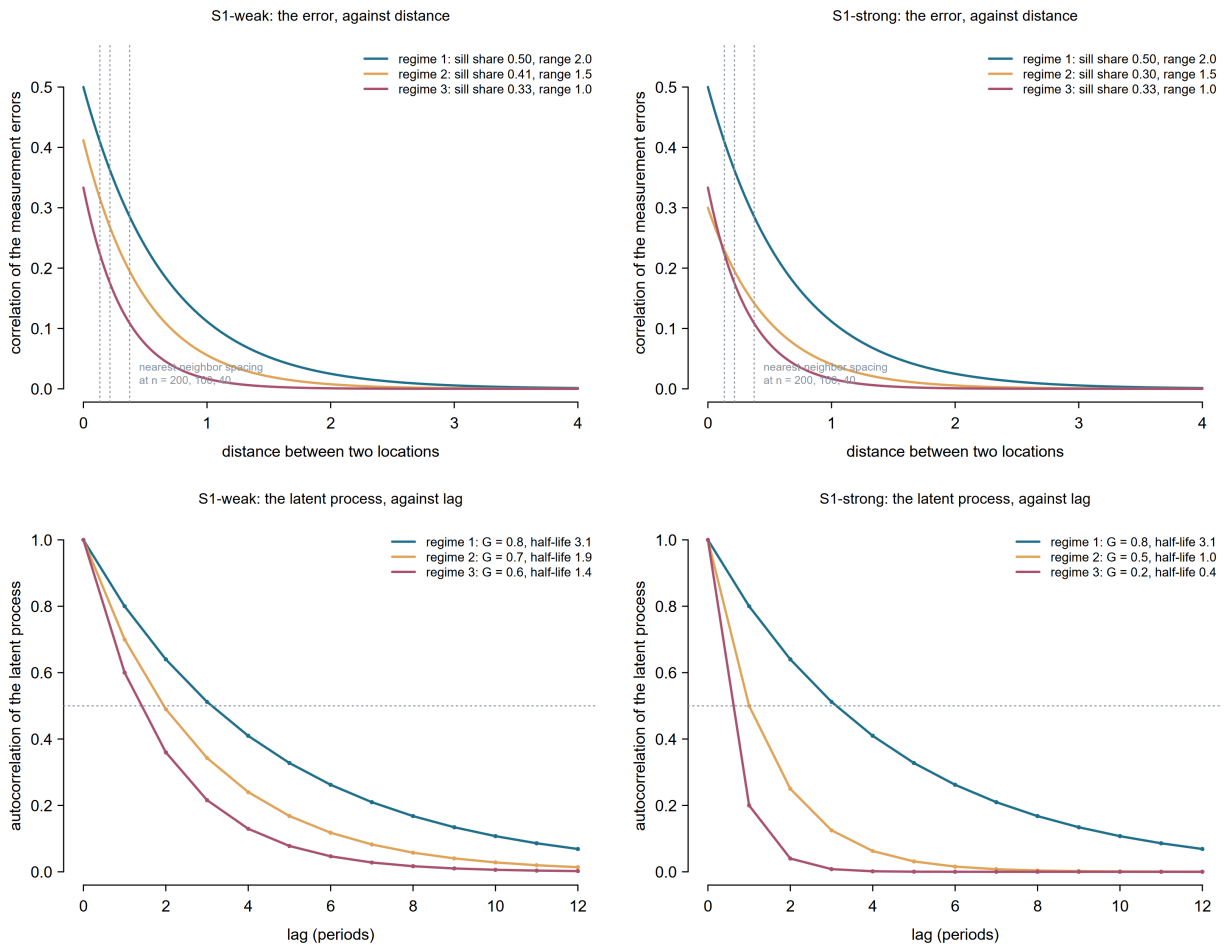


Figure B3: Top: the correlation of the measurement errors of two locations of the same regime against their distance, $\sigma_\omega^2/(\sigma_\epsilon^2+\sigma_\omega^2)\, e^{-\theta h}$; the dotted lines mark the median distance between nearest neighbors at $d = 200$, 100 and 40 in the reference geometry. Bottom: the autocorrelation of the latent process against the lag, G^h; the dotted line marks one half, and the half-life $\log 0.5/\log G$ is given in the legend. Left: S1-weak; right: S1-strong.

The spatial parameters of the third regime. The values of $(\sigma_\epsilon^2, \sigma_\omega^2, R)$ of regime 3 of S1-strong were chosen so that its spatial parameters are identified at the sizes of the design. A first calibration gave regime 3 a partial sill of one sixth of its error variance and a range of half a unit, about three times the median distance between nearest neighbors at $d = 100$. Its spatial component was then too weak and too short to be told apart from the nugget: in pilot replications the estimated range of that regime was long, and a field spanning the whole regime absorbed part of its almost white latent process ($G = 0.2$), since a field nearly constant in space and independent over time is indistinguishable from a latent process with no persistence. With a partial sill of one third of the error variance and a range of one unit the estimates of the regime center on the true values. The total error variance $s_3^2 = 0.60$, which is all the assignment step sees of the error, is unchanged, so the separation of the regimes (Section B4) is the same.

Two examples. Figure B4 shows one data set at each level of separation, in time and in space. With independent regimes the latent paths of the regimes move on their own; with shared shocks

they move together, and with weak contrasts the regimes differ little both in the series and in the maps.

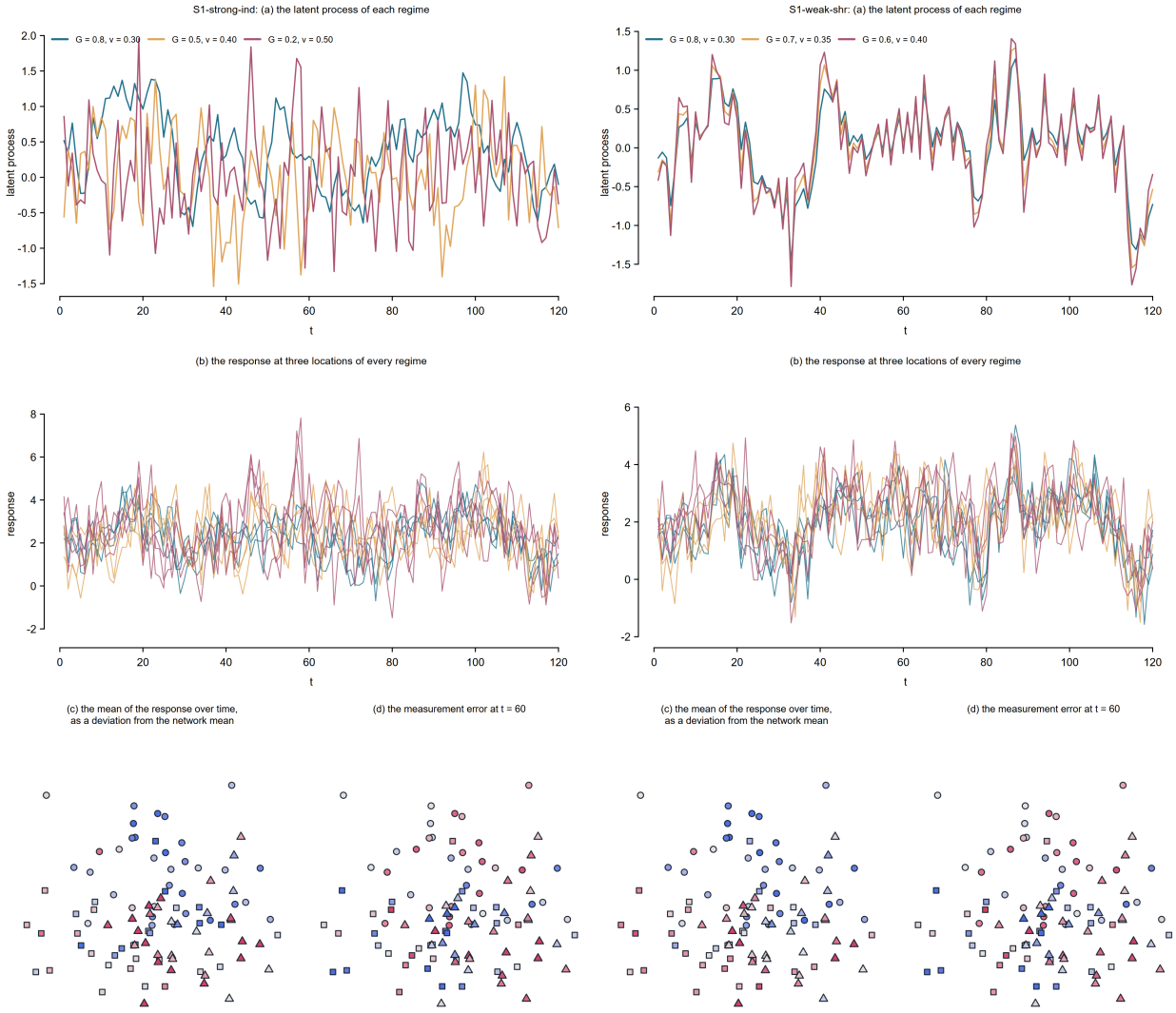


Figure B4: One data set of S1-strong in the variant `ind` (left) and of S1-weak in the variant `shr` (right), at the reference cell ($d = 100$, $T = 120$, medium-high overlap), first replication. (a) The latent process of each regime: independent in `ind`, driven by the same shocks in `shr`. (b) The response at three locations of each regime. (c) The mean of the response over time at every location, as a deviation from the mean over the network (blue below, red above); circles, squares and triangles mark regimes 1, 2 and 3. (d) The measurement error at $t = 60$, on the same scale for the two data sets.

B3.3 The variants: dependence between regimes

The generator has two links between regimes, and they are separate assumptions on different components of the model: ρ governs whether the latent processes share their shocks, and the construction of the error governs whether one field crosses the borders between regimes. They are

paired by a choice of design, so that a scenario has an "all independent" and an "all shared" variant (Table B3).

Table B3: The variants of dependence between regimes.

code	ρ	measurement error	exists when
`ind`	0	fields by regime	always, except in the pooled scenario
`shr`	1	one field if θ is common, otherwise fields by regime	always

The variant `ind` is exactly the SC-STEM model: independent latent paths and a block-diagonal error covariance. The variant `shr` shares everything the scenario permits. With $\rho = 1$ the regimes are driven by one sequence of shocks ξ_t, which each regime filters with its own persistence G_k and scales by its own $\sigma_{\eta k}$, so that the correlation of two latent paths is κ_{kl} of (B7). Where the ranges differ, only the latent process can be shared, since one field has one range; where θ is common, the latent process and the field are both shared.

What SC-STEM makes of the shared variant. Under `shr` the law of each regime taken alone is still exactly its STEM model: a Gaussian vector restricted to a subset of its components is Gaussian with the corresponding sub-covariance, and each $y^{(k)}$ is marginally an AR(1) with parameters $(G_k, \sigma^2_{\eta k})$. SC-STEM is wrong only in assuming that the regimes are independent. Given the partition, the sum of the regime log-likelihoods becomes a composite likelihood, the product of the block marginals of the true joint law, and each $\hat{\Psi}_k$ remains consistent. The refit-with-clustering bootstrap regenerates the regimes independently, and so ignores the correlation between the estimates of different regimes that the shared variant induces, which concerns the tests of the between-regime differences; and the information criteria compare the sum of the regime log-likelihoods across K, a sum that ignores the dependence between the blocks. The independence of the regimes, however, is itself information about the partition (Section B4): sharing the latent process weakens the channel through the smoothed latent paths that the assignment score uses, while sharing the field changes nothing in the score, which does not see spatial correlation, and acts only through the exact likelihood of the parameter step and through the comparison across K. The shared variant thus tests the robustness of SC-STEM to a violation of its assumptions that leaves each regime correctly specified.

Two empirical readings. In the independent variant the border between regimes is a barrier: the regimes are distinct systems with their own drivers, as river basins separated by a watershed divide, or hydraulically disconnected aquifers. In the shared variant the border is a change of response, not a barrier: all regimes are subject to the same large-scale driver and filter it with their own persistence and amplitude, as temperature anomalies on a coast and inland driven by the same synoptic weather, or regional labor markets responding to the same national cycle. The second is the more common case in real data.

The pooled likelihood against the likelihood of a partition. When the data come from the pooled STEM model, the SC-STEM log-likelihood of a partition into $K > 1$ regimes is the sum of block marginals, while the pooled log-likelihood is the joint. At the true parameters,

$$\mathbb{E}\Big[\log p(z) - \sum_k \log p(z_{\mathcal{C}_k})\Big] = \mathcal{I}\big(z_{\mathcal{C}_1}; \ldots; z_{\mathcal{C}_K}\big) \;\geq\; 0, \tag{B10}$$

the multi-information between the blocks, which is zero only if the blocks are independent. Splitting a pooled network into regimes therefore loses likelihood, besides paying the penalty of the criterion, and the loss grows with T and with the strength of the dependence that the blocks share: the variance of the common latent process and the range of the common field. The false-positive rate measured in the pooled scenario is therefore a function of its latent variance and range, which are those of the baseline regime.

B3.4 The nine scenario-variants

The study runs nine scenario-variants (Table B4): S2 (question Q2); S1-strong-`ind`, the reference case, in which the data are generated by the SC-STEM model written in full; S1-strong-`shr` and S1-weak-`shr`, the two levels of separation with coupled regimes (Q1); and the five S3-x-`shr`, the contribution of each block in the variant where the separation rests on the parameters (Q3). The contrasts in the parameters are studied in the variant `shr` because in the variant `ind` the independent latent paths separate the regimes on their own, whatever the parameters (Proposition B1).

Table B4: The scenario-variants and their relation to the data-generating processes of SC-STEM and STEM.

scenario	variants	generated by SC-STEM?	by STEM?
S2	one	yes, at $K = 1$	yes
S1-strong	`ind`	yes, exactly: the reference case	no
S1-strong, S1-weak	`shr`	each regime exactly; the latent paths are correlated across regimes	no
S3-beta, -G, -Seta	`shr`	each regime exactly; latent paths correlated	no
S3-theta, -error	`shr`	each regime exactly; latent paths and spatial field shared	no

B4 The separation of the regimes

The separation of the regimes is induced by design through the parameter values of the scenarios (Section B3) and the dependence between regimes of the variants (Section B3.3), that were fixed to be interpretable: together they decide how much a single location tells about the regime it belongs to, and thus how hard the partition is to recover. Here we quantify the degree of separation they induce by the expected log-likelihood ratio that a location accumulates in favor of its own regime against the nearest wrong one, when the assignment step of the algorithm scores it (Section 2.3 of the manuscript). The measure is therefore a by-product of the design, not a target for it: no parameter value was tuned to reach a given value of the measure. The measure serves three purposes.

1. It orders the scenario-variants from the most to the least separated (the order of Figure 1 of the manuscript and of the tables of Section B6);
2. It identifies, before the experiment is run, the hard cases of the design (i.e., S1w-`shr` and S3beta-`shr`)
3. It explains the results: the order it predicts is the order observed in the recovery of the number of regimes and of the partition (Section B6, and Section 4.2 of the manuscript), and, since the assignment step does not see the spatial range, a common range (S3theta) leaves the procedure close to S1s-`shr`.

B4.1 What each step of the algorithm sees

Given the partition, the parameter step fits each regime by the EM algorithm on its exact likelihood, which involves every component of Ψ_k. Given the parameters, the assignment step scores location i against regime l by the pseudo-likelihood of Section 2.3 of the manuscript,

$$S_i(l) = \sum_{t=1}^{T} \left[-\tfrac{1}{2} \log\bigl(2\pi s_l^2\bigr) - \frac{\bigl(z_{it} - x_{it}^\top \beta_l - \hat{y}_t^{(l)}\bigr)^2}{2 s_l^2} \right], \qquad s_l^2 = \sigma_{\epsilon l}^2 + \sigma_{\omega l}^2, \tag{B11}$$

with $\hat{y}_t^{(l)}$ the smoothed latent path of regime l. The score (B11) sees β_l, the total error variance s_l^2, and the dynamics $(G_l, \sigma_{\eta l}^2)$ only through the smoothed path. It sees neither the range θ_l nor the split of s_l^2 between nugget and partial sill. A contrast in θ therefore separates the regimes much less, for the assignment step, than a contrast of the same nominal size in β. Moreover, the score (B11) is a sum over time of one term per period: it treats the periods as conditionally independent given the latent path, as the error of STEM is white in time.

B4.2 The quantity

Take a location i of regime k and a wrong regime $l \neq k$. Ignoring the Potts term, the assignment step prefers k to l when $S_i(k) - S_i(l) > 0$. The measure of separation is the expected value of that difference, that is,

$$\mathbb{E}\bigl[S_i(k) - S_i(l)\bigr] = \sum_{t=1}^{T} \mathbb{E}\bigl[d_{it}\bigr], \qquad d_{it} = \log \varphi\bigl(z_{it};\, m_{it}^{(k)},\, s_k^2\bigr) - \log \varphi\bigl(z_{it};\, m_{it}^{(l)},\, s_l^2\bigr), \tag{B12}$$

where $\varphi(\cdot\,; m, s^2)$ is the normal density and $m_{it}^{(r)} = x_{it}^\top \beta_r + \hat{y}_t^{(r)}$ is the mean that the score attributes to location i under regime r.

Let us denote D_{kl} the expectation of one of those terms. By linearity of expectation, and because under (A2)–(A4) every period has the same expectation,

$$\mathbb{E}\bigl[S_i(k) - S_i(l)\bigr] = \sum_{t=1}^{T} \mathbb{E}[d_{it}] = T\, D_{kl}. \tag{B13}$$

TD_{kl} is in units of log-likelihood (nats): it is the expected log-likelihood ratio in favor of the true regime that one location accumulates over its whole series, and $e^{TD_{kl}}$ the corresponding likelihood ratio. It grows linearly in T, while its standard deviation grows more slowly, so a location is assigned more reliably the longer its series. Notice that this step uses only linearity, it does not require the terms to be independent over time (they are not independent, since the covariate and the latent paths are autocorrelated).

B4.3 Assumptions

(A1) *Oracle paths and parameters.* The parameters of every regime are at their true values and the smoothed paths equal the true ones, $\hat{y}_t^{(r)} = y_t^{(r)}$. This is the situation the algorithm reaches when the regimes are well estimated; it is an idealization, and the measure is therefore the separation available to the assignment step at best.

(A2) *Covariate.* x_{it} is independent of the latent processes and of the errors, with $\mathbb{E}x_{it} = 0$ and $\mathbb{E}x_{it}^2 = 1$; the covariate is standardized over all its dT values, so these hold on average over the network rather than exactly at each location.

(A3) *Stationarity.* The latent processes are at their stationary distribution, with the moments (B7).

(A4) *Errors.* e_{it} has mean zero and variance s_k^2, and is independent of the covariate and of the latent processes.

B4.4 Derivation

Under the true regime the residual is the error itself, $z_{it} - m_{it}^{(k)} = e_{it}$, by (A1). Under the wrong regime,

$$z_{it} - m_{it}^{(l)} = \underbrace{x_{it}^\top(\beta_k - \beta_l)}_{a_{it}} + \underbrace{y_t^{(k)} - y_t^{(l)}}_{b_t} + e_{it}. \tag{B14}$$

Writing out the two log densities,

$$d_{it} = -\tfrac{1}{2}\log s_k^2 - \frac{e_{it}^2}{2s_k^2} + \tfrac{1}{2}\log s_l^2 + \frac{(a_{it}+b_t+e_{it})^2}{2s_l^2}.$$

Taking expectations, $\mathbb{E}e_{it}^2 = s_k^2$ by (A4), and $\mathbb{E}(a_{it}+b_t+e_{it})^2 = \mathbb{E}a_{it}^2 + \mathbb{E}b_t^2 + \mathbb{E}e_{it}^2$, because the three cross products have zero expectation: $\mathbb{E}[a_{it}e_{it}] = 0$ and $\mathbb{E}[b_t e_{it}] = 0$ by (A4), and $\mathbb{E}[a_{it}b_t] = \mathbb{E}[a_{it}]\,\mathbb{E}[b_t] = 0$ by (A2) and $\mathbb{E}b_t = 0$. Hence

$$D_{kl} := \mathbb{E}\big[d_{it}\big] = \frac{1}{2}\left[\frac{s_k^2 + \Delta_{kl}^2}{s_l^2} - 1 - \log\frac{s_k^2}{s_l^2}\right], \qquad \Delta_{kl}^2 = \mathbb{E}a_{it}^2 + \mathbb{E}b_t^2. \tag{B15}$$

The two components of Δ_{kl}^2 follow from (A2) and (A3). With $x_{it} = (1, x_{it})^\top$ and $\Delta\beta_j = \beta_{jk} - \beta_{jl}$, $\mathbb{E}a_{it}^2 = \Delta\beta_0^2 + 2\Delta\beta_0\Delta\beta_1\mathbb{E}x_{it} + \Delta\beta_1^2\mathbb{E}x_{it}^2 = \Delta\beta_0^2 + \Delta\beta_1^2$; and from (B7), $\mathbb{E}b_t^2 = v_k + v_l - 2\,\mathrm{Cov}(y_t^{(k)}, y_t^{(l)})$, with $\mathrm{Cov}(y_t^{(k)}, y_t^{(l)}) = \sum_{j\geq 0}(G_kG_l)^j\rho\,\sigma_{\eta k}\sigma_{\eta l} = \rho\,\sigma_{\eta k}\sigma_{\eta l}/(1 - G_kG_l)$. Therefore

$$\Delta_{kl}^2 = \underbrace{(\beta_{0k} - \beta_{0l})^2 + (\beta_{1k} - \beta_{1l})^2}_{\text{mean term}} + \underbrace{v_k + v_l - 2\rho\,\kappa_{kl}\sqrt{v_k v_l}}_{\text{latent term, } \mathbb{E}b_t^2}. \tag{B16}$$

Reading. D_{kl} is the expected Kullback–Leibler divergence between the two normal densities that the score compares at one time point,

$$D_{kl} = \mathbb{E}_{x,y}\Big[\mathrm{KL}\big(N(m_{it}^{(k)}, s_k^2) \;\|\; N(m_{it}^{(l)}, s_l^2)\big)\Big],$$

since $\mathrm{KL}(N(\mu_1,\sigma_1^2)\,\|\,N(\mu_2,\sigma_2^2)) = \tfrac{1}{2}[\sigma_1^2/\sigma_2^2 + (\mu_1-\mu_2)^2/\sigma_2^2 - 1 - \log(\sigma_1^2/\sigma_2^2)]$ and $\mathbb{E}(m_{it}^{(k)} - m_{it}^{(l)})^2 = \Delta_{kl}^2$. It is non-negative, zero if and only if $s_k^2 = s_l^2$ and $\Delta_{kl}^2 = 0$, and not symmetric in k and l. It separates into a mean part, $\Delta_{kl}^2/(2s_l^2)$, and a variance part, $\tfrac{1}{2}[s_k^2/s_l^2 - 1 - \log(s_k^2/s_l^2)]$, which is the only place where the error variances enter.

B4.5 Formal consequences

Proposition B1 (Independence alone separates the regimes)**.** *Under $\rho = 0$, $\mathbb{E}b_t^2 = v_k + v_l$ whatever the parameters, so*

$$D_{kl} \;\geq\; \frac{1}{2}\left[\frac{s_k^2 + v_k + v_l}{s_l^2} - 1 - \log\frac{s_k^2}{s_l^2}\right] \;\geq\; \frac{v_k + v_l}{2s_l^2} \quad \textit{when } s_k^2 = s_l^2.$$

With independent regimes the latent paths separate the regimes on their own, with a bound set by the weight of the latent signal relative to the noise and not by any contrast in the parameters. In the variant `ind`, therefore, the level of separation of the parameters governs the estimation of Ψ_k and the choice of K, but hardly the recovery of the partition, which the latent channel already secures. This is coherent with `ind` being the reference case: the model is correct and the problem is easy.

Proposition B2 (Decomposition of the latent term)**.** $\mathbb{E}b_t^2 = \left(\sqrt{v_k} - \sqrt{v_l}\right)^2 + 2\sqrt{v_k v_l}\left(1 - \rho\,\kappa_{kl}\right)$.

The first term is the difference in amplitude, the second the difference in timing, which is due to independent shocks ($\rho < 1$) or to different persistence ($\kappa_{kl} < 1$). With $\rho = 1$ and a common G the second term vanishes and only the amplitude separates the paths, which are then perfectly correlated and differ only in scale. With $\rho = 1$, a common G and a common v the latent term vanishes altogether, and the separation rests on β and s^2 alone.

Proposition B3 (What the measure does not see)**.** *D_{kl} depends on the error only through the total variances s_k^2 and s_l^2. It does not depend on θ, on the split of s^2 into nugget and partial sill, or on whether the error is drawn by regime or as one field.*

In particular S3theta has the same measure as S1-strong, since the two differ only in θ: whatever the range contributes to the recovery of the regimes, it contributes through the exact likelihood of the parameter step, not through the assignment step.

B4.6 Limits of the measure

The measure describes the assignment step at the oracle: in practice $\hat{y}^{(l)}$ is estimated from the locations currently in regime l, which early in the alternation include misassigned ones, and the parameters are estimates. It ignores the Potts term, which adds to the score of a label the agreement with the neighbors. It is pairwise and per location. It measures what the assignment step can use, not the total information in the data, which the parameter step reads through the exact likelihood. It quantifies the separation that the design induces and orders the scenario-variants by it; it was not used to calibrate them, and it is not a prediction of the recovery rates.

B4.7 Empirical values by scenario-variants

Table B5 evaluates (B15)–(B16) on the nine scenario-variants. In S1-strong-`ind` the latent term alone is 0.70, larger than any mean term of the design, and $120 \min D$ is 107 nats. In the variant `shr` the latent term nearly vanishes and the separation rests on the parameters: S1-weak is poorly separated, about 4.5 nats per location over 120 periods, a margin that the Potts term and the neighbors can overturn, and S1-strong is well separated, about 25 nats. Against S1-strong, S3-beta falls to 6.6 nats, since the coefficients are the main separator for the assignment step; S3-G falls to 19.5, since with a common persistence and common shocks the latent paths differ only in amplitude; S3-Seta falls to 20.6, since the amplitudes become smaller; S3-theta stays at 24.6, since the score does not see the range, so that whatever S3-theta loses in recovery is due to the parameter step alone; and S3-error rises to 35.5, since the common error variance is the smallest of S1-strong, so every mean contrast is measured against a smaller noise. S1w-`shr` and S3beta-`shr` are therefore the two challenging settings of the design. In S1w every contrast is small. In S3beta the common coefficients remove the main lever of the assignment step, since what remains (i.e., the dynamics, the error variance and the range) the score sees weakly or not at all. A margin of a few nats per location can be overturned by the Potts term and by the neighbors, and these are the two scenario-variants

in which the procedure is expected to under-select the number of regimes on small networks and short series.

Table B5: The separation measure D_{kl} per time point for the ordered adjacent pairs; at the hardest pair, its mean term $(\Delta\beta_0)^2 + (\Delta\beta_1)^2$ and its latent term $\mathbb{E}b_i^2$; and the expected log-likelihood ratio $T \min D$ accumulated by one location over $T = 60$, 120 and 365 periods, for the nine scenario-variants of the study. S2 has $D = 0$ by construction.

scenario-variant	ρ	D_{12}	D_{21}	D_{23}	D_{32}	min D	mean	latent	60 min D	120 min D	365 min D
`S1s-ind`	0	0.892	1.113	0.908	1.089	0.892	0.18	0.700	53.5	107.0	325.4
`S1s-shr`	1	0.292	0.363	0.205	0.246	0.205	0.18	0.057	12.3	24.6	74.9
`S1w-shr`	1	0.047	0.050	0.037	0.040	0.037	0.02	0.013	2.2	4.5	13.6
`S3beta-shr`	1	0.112	0.138	0.055	0.066	0.055	0.00	0.057	3.3	6.6	20.1
`S3G-shr`	1	0.199	0.247	0.162	0.194	0.162	0.18	0.006	9.7	19.5	59.3
`S3Seta-shr`	1	0.276	0.343	0.172	0.205	0.172	0.18	0.017	10.3	20.6	62.6
`S3theta-shr`	1	0.292	0.363	0.205	0.246	0.205	0.18	0.057	12.3	24.6	74.9
`S3error-shr`	1	0.350	0.350	0.296	0.296	0.296	0.18	0.057	17.8	35.5	108.0

B5 The design of the experiment

Factors and blocks. The nine scenario-variants are crossed with five margins, whose codes also name the cells of the design (the scripts call the number of locations `n`):

`n` the number of locations, $d \in \{40, 100, 200, 400\}$;

`T` the length of the series, $T \in \{60, 120, 365\}$;

`w` the geometry of the regimes in space, through the distance ω between their centers (Section B2.2): `w0.00` total overlap, the regimes occupying the same area, so that space does not help; `w0.70` medium-high overlap, contiguous regimes interpenetrating along their borders, the reference; `w1.00b` medium-low overlap, the centers farther apart with every regime keeping its spread, so that the network widens; `w1.00` strong separation, the centers farther apart with more compact regimes, at a fixed spread of the network;

`bal, unb` regimes of equal size, or in the proportions 1:2:3;

`knn` the number of neighbors $m \in \{3, 5, 10\}$ of the graph of the Potts penalty.

The reference cell is $d = 100$, $T = 120$, $\omega = 0.7$, balanced regimes and $m = 5$. The design is organized in blocks (Table B6): each block is the full factorial of the margins it lists, with every other margin at its reference level, crossed with the scenario-variants, and the design is the union of the blocks, a cell shared by two blocks being run once. The core crosses d, T and three geometries (225 cells); the block *spread* adds the fourth geometry (72); the blocks *balance* (8), *knn* (18) and *n400* (9) vary one margin at the reference cell; 332 cells in all. In S2 the geometry and the balance are vacuous, and the scenario enters once per (d, T, m). No network smaller than $d = 40$ is used: the grid reaches $K = 4$ regimes of at least six locations each, and $d = 40$ gives about 13 locations per regime.

Figure B5 shows how the settings differ, on one replication at $d = 100$. The top row is the geometry: the locations of the three regimes coincide in space under total overlap, interpenetrate along their borders at the reference, and separate as the centers move apart, while the unbalanced design gives the regimes 17, 33 and 50 locations. The other panels are the scenario-variants at the reference cell: a single regime in S2; three regimes moving independently in S1s-`ind`; in the `shr`

Table B6: The blocks of the design. A cell is counted under the first block that contains it; the reference cell belongs to the core.

block	margins (the others at the reference)	new cells
core	$n \in \{40, 100, 200\}$, $T \in \{60, 120, 365\}$, $\omega \in \{0, 0.7, 1\}$	225
spread	n and T as the core, $\omega = 1$ with the spread of a regime fixed	72
balance	regime sizes $1:2:3$	8
knn	$m \in \{3, 10\}$ neighbors	18
n400	$n = 400$	9
total		332

variants three regimes driven by the same shocks, which differ by level, amplitude and persistence, by much in S1s-`shr` and by little in S1w-`shr`, and whose mean levels coincide when the regression coefficients are common (S3beta-`shr`).

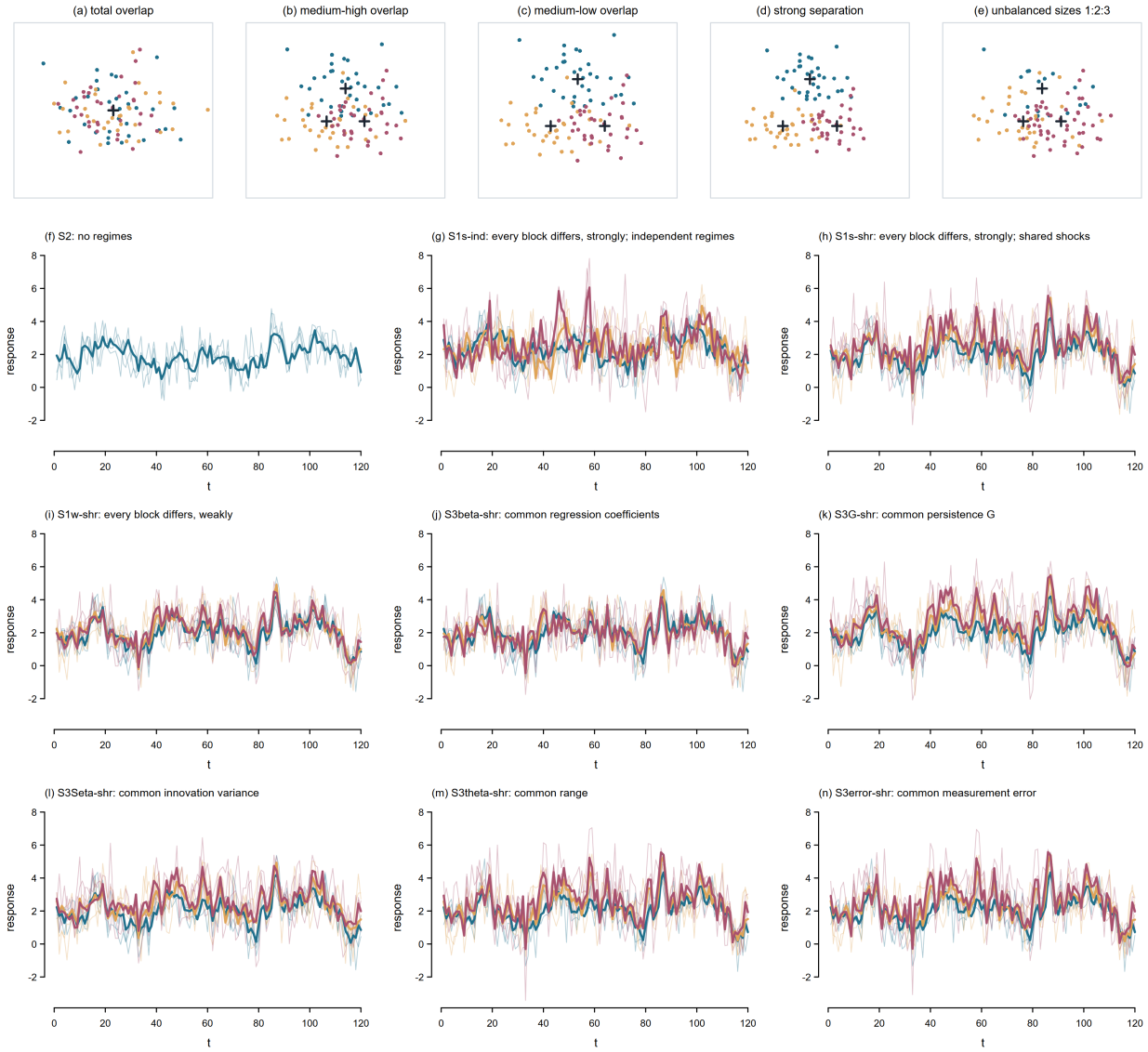


Figure B5: The settings of the study at $d = 100$ and $T = 120$, first replication. (a)–(e) The locations, colored by regime, in the four geometries and with regime sizes 1:2:3; the crosses mark the centers of the regimes. (f)–(n) The response of every scenario-variant: three locations of each regime (thin lines) and the mean of the response over the locations of each regime (thick lines).

How the blocks are read. The core, with its fourth geometry in the block *spread*, maps the performance of the procedure along the three factors that govern the information available to it: d, how many locations each regime has; T, how much evidence accumulates for each label; and the geometry, whether and how much space helps. The other three blocks are checks of robustness at the reference cell, each with an expected direction.

Balance. At 1:2:3 and $d = 100$ the regimes have 17, 33 and 50 locations. The likelihood gain of a regime grows with its size, so the smallest regime carries half the evidence of a balanced one, and its border locations have more neighbors in the larger regimes, toward which the Potts term pulls them. Expected: no change in the well-separated scenario-variants; in the hard ones the

small regime may be absorbed and K underselected.

Graph. The graph of the penalty is a choice of the analyst, and under the automatic scale the factor c divides by the mean degree, so raising m spreads the same total pull over more and longer edges rather than increasing it (Section A3 of Computational and Algorithmic Supplement A). Expected: results nearly invariant to m; a sparse graph is more local, a dense one smooths across the borders over longer edges.

National scale. $d = 400$ is the size of a national monitoring network, and the scale at which the theory of Section A6 of Computational and Algorithmic Supplement A predicts a risk: at $T = 120$ and $q = 8$ parameters per regime, the gain that a superfluous fourth regime can extract from the search over partitions is bounded by $2d\log(4/3)$, which exceeds the BIC increment $q\log(dT)$ from $d = 200$ on (115 against 81 at $d = 200$, 230 against 86 at $d = 400$). The bound is conservative, since neighboring partitions share most of their labels. The block, with $d = 200$ in the core, says whether the BIC over-selects on wide networks in practice.

Varying one margin at a time does not show interactions, so the results of these three blocks are statements of robustness at the reference cell and not in general (Morris et al., 2019).

What is fitted. Every replication fits the grid of $K = 1, \ldots, 4$ regimes and of the penalties $\phi = 0$, 0.025, 0.05, 0.1, 0.2, 0.5 and 1, and applies the two-step selection rule with its default band $[0.025, 0.2]$ (Section 3.2 of the manuscript). Every penalized fit starts from the unpenalized fit at the same K, which also fixes the scale of its penalty. The selection is read off the selected fit. The recovery of the partition and of the parameters is read off the fit at the true number of regimes and $\phi = 0.05$, so that a poor estimate is not confounded with a poor selection; the pooled fit is the benchmark. All three are members of the grid. Every fit runs the EM algorithm of Section A1 of Computational and Algorithmic Supplement A, accelerated by SQUAREM in the pooled fit and in the final refits and run plainly inside the alternation, with the stopping rule of Section A1.4 of Computational and Algorithmic Supplement A.

B6 Extended results

The results below come from 200 replications of every cell. Seventeen replications failed for reasons unrelated to the estimator and are left out. Shares are given with their Monte Carlo standard errors in parentheses, and distributions over the replications as boxplots, whose boxes span the quartiles and whose whiskers reach 1.5 times the interquartile range. K_0 denotes the true number of regimes.

B6.1 The reference cell

Table B7 gives every scenario-variant at the reference cell. The seven well-separated scenario-variants select the three regimes in every replication and recover the partition almost exactly, and their fitted signal has a root mean squared error between a quarter and two fifths of that of the pooled fit. S1w-`shr` and S3beta-`shr` select three regimes in about a quarter of the replications, two otherwise, and at K_0 recover the partition with an ARI of 0.59 and 0.66. S2 is never split.

Table B7: The reference cell ($d = 100$, $T = 120$, medium-high overlap, balanced sizes, $m = 5$). M: replications; $\Pr(\hat{K} = K)$: share selecting the true number of regimes, with its standard error; ARI at K and at $\hat{K}$: mean adjusted Rand index of the partition at the true and at the selected number of regimes; modal $\hat{\phi}$: the penalty selected most often when $\hat{K} = K$; RMSE ratio: median ratio of the root mean squared error of the fitted signal at K_0 to that of the pooled fit.

scenario-variant	M	$\Pr(\hat{K} = K_0)$	ARI at K_0	ARI at $\hat{K}$	modal $\hat{\phi}$	RMSE ratio
`S2`	200	1.00 (0.00)	—	—	0.000	1.00
`S1s-ind`	200	1.00 (0.00)	1.00 (0.00)	1.00	0.000	0.24
`S1s-shr`	199	1.00 (0.00)	1.00 (0.00)	1.00	0.000	0.36
`S1w-shr`	200	0.25 (0.03)	0.59 (0.01)	0.57	0.200	0.91
`S3beta-shr`	200	0.22 (0.03)	0.66 (0.01)	0.63	0.025	0.75
`S3G-shr`	200	0.99 (0.00)	1.00 (0.00)	1.00	0.000	0.42
`S3Seta-shr`	199	1.00 (0.00)	1.00 (0.00)	1.00	0.000	0.37
`S3theta-shr`	200	1.00 (0.00)	1.00 (0.00)	1.00	0.000	0.42
`S3error-shr`	200	1.00 (0.00)	1.00 (0.00)	1.00	0.000	0.41

B6.2 Selection of the number of regimes

Table B8 gives the share of the replications that select the true number of regimes in every cell of the core, and Figure B6 the selection error $\hat{K} - K_0$ of every replication, which also shows the direction of the errors.

- In the well-separated scenario-variants the share is at least 0.82 in every cell with $d \geq 100$, and at least 0.99 from $T = 120$ on. At $d = 40$ it depends on T: on average over these scenario-variants and the geometries, 0.54 at $T = 60$, 0.83 at $T = 120$ and 0.98 at $T = 365$. S1s-`ind` is identified in practically every cell.
- In S1w-`shr` and S3beta-`shr` the share is close to zero at $d = 40$ and grows with d and T, to between 0.61 and 0.94 at $d = 200$ and $T = 365$.
- The geometry acts in the same direction in every scenario-variant: the more the regimes are separated in space, the less often their number is identified. At $d = 40$ and $T = 60$ the well-separated scenario-variants are identified in 71% of the replications when the regimes overlap completely and in 36% when they are strongly separated; the medium-low overlap, in which the regimes keep the spread of the reference cell, lies between the two.
- The errors are mostly under-selections. Over-selection appears where the weak regimes overlap in space on the larger networks: at $d = 200$ with total or medium-high overlap, S1w-`shr` and S3beta-`shr` select four regimes in up to 27% of the replications, against at most 7% with the medium-low overlap and never with strong separation; in the well-separated scenario-variants it stays below 10% in every cell, and at most 1% from $T = 120$ on. This is the over-selection that the search over partitions can produce on wide networks (Section 3.2 of the manuscript). In S2 (Table B9) no replication of any cell selects more than one regime.

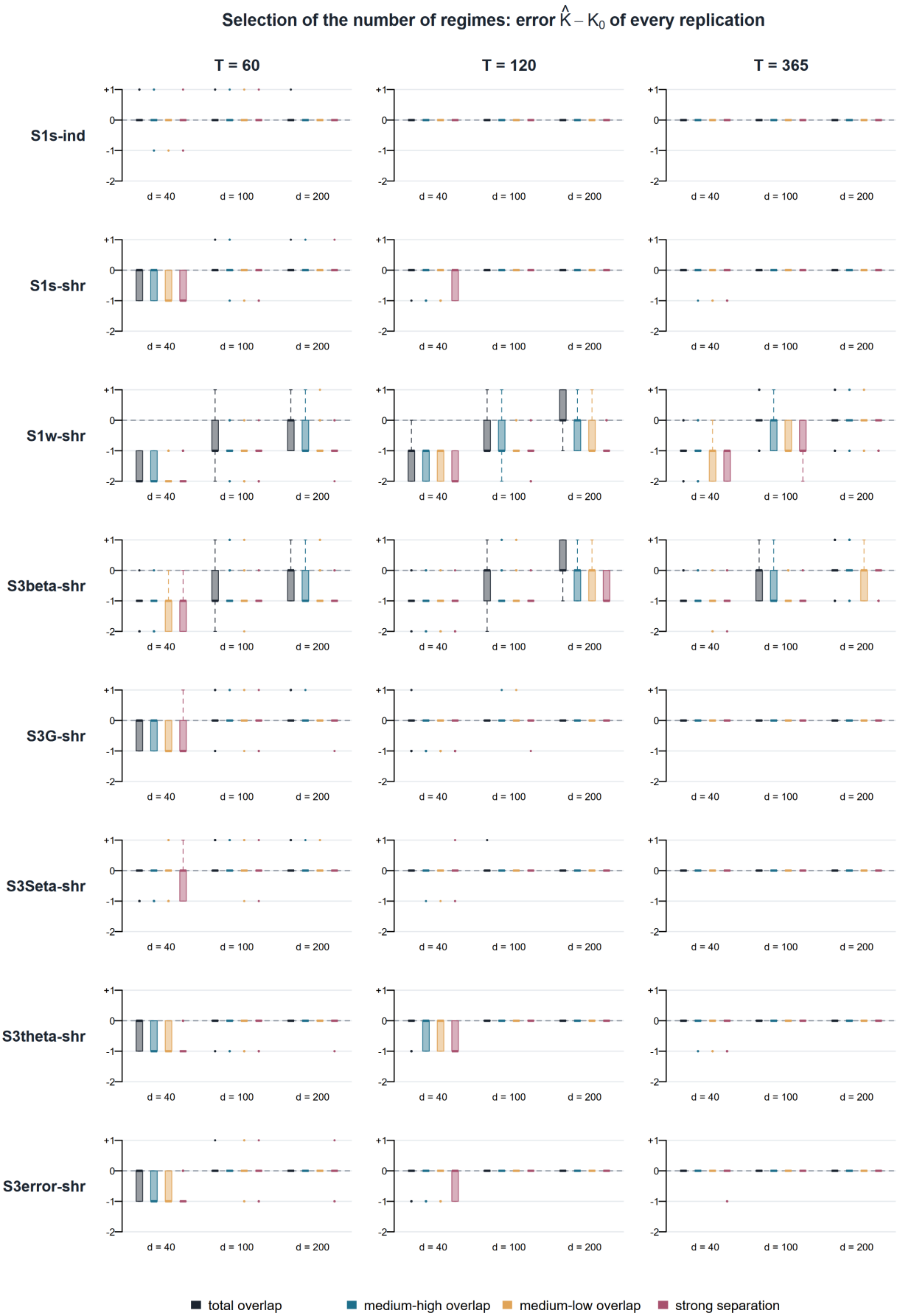


Figure B6: Selection of the number of regimes in the core of the design: boxplots of the selection error $\hat{K} - K_0$ over the replications of every cell (0: the true number; -1: one regime too few; $+1$:

Table B8: Share of the replications selecting the true number of regimes in the core of the design, by scenario-variant, geometry, T and d.

		$T = 60$			$T = 120$			$T = 365$		
scenario-variant	geometry	$d = 40$	$d = 100$	$d = 200$	$d = 40$	$d = 100$	$d = 200$	$d = 40$	$d = 100$	$d = 200$
`S1s-ind`	total overlap	0.98	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00
	medium-high overlap	0.98	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	medium-low overlap	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	strong separation	0.98	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
`S1s-shr`	total overlap	0.74	0.99	0.99	0.99	1.00	1.00	1.00	1.00	1.00
	medium-high overlap	0.59	0.98	0.99	0.94	1.00	1.00	0.99	1.00	1.00
	medium-low overlap	0.48	0.98	1.00	0.82	1.00	1.00	0.98	1.00	1.00
	strong separation	0.31	0.94	0.99	0.61	1.00	1.00	0.95	1.00	1.00
`S1w-shr`	total overlap	0.00	0.22	0.43	0.01	0.32	0.56	0.07	0.78	0.82
	medium-high overlap	0.00	0.10	0.29	0.00	0.25	0.60	0.06	0.68	0.94
	medium-low overlap	0.00	0.03	0.10	0.00	0.09	0.42	0.01	0.43	0.94
	strong separation	0.00	0.00	0.05	0.00	0.04	0.20	0.00	0.28	0.83
`S3beta-shr`	total overlap	0.01	0.28	0.46	0.04	0.44	0.52	0.14	0.65	0.67
	medium-high overlap	0.01	0.15	0.36	0.06	0.22	0.59	0.10	0.47	0.66
	medium-low overlap	0.00	0.10	0.19	0.03	0.10	0.50	0.06	0.13	0.61
	strong separation	0.01	0.02	0.10	0.01	0.03	0.33	0.04	0.02	0.78
`S3G-shr`	total overlap	0.72	0.95	0.94	0.94	1.00	1.00	1.00	1.00	1.00
	medium-high overlap	0.63	0.98	0.99	0.94	0.99	1.00	1.00	1.00	1.00
	medium-low overlap	0.47	0.98	1.00	0.91	0.99	1.00	1.00	1.00	1.00
	strong separation	0.38	0.96	0.99	0.83	0.99	1.00	1.00	1.00	1.00
`S3Seta-shr`	total overlap	0.89	0.92	0.94	1.00	0.99	1.00	1.00	1.00	1.00
	medium-high overlap	0.82	0.98	0.99	0.99	1.00	1.00	1.00	1.00	1.00
	medium-low overlap	0.74	0.98	0.99	0.99	1.00	1.00	1.00	1.00	1.00
	strong separation	0.66	0.99	1.00	0.98	1.00	1.00	1.00	1.00	1.00
`S3theta-shr`	total overlap	0.54	0.98	1.00	0.85	1.00	1.00	1.00	1.00	1.00
	medium-high overlap	0.38	0.96	1.00	0.71	1.00	1.00	0.98	1.00	1.00
	medium-low overlap	0.35	0.96	1.00	0.64	1.00	1.00	0.96	1.00	1.00
	strong separation	0.20	0.82	0.99	0.40	1.00	1.00	0.84	1.00	1.00
`S3error-shr`	total overlap	0.67	0.99	1.00	0.94	1.00	1.00	1.00	1.00	1.00
	medium-high overlap	0.48	1.00	1.00	0.85	1.00	1.00	1.00	1.00	1.00
	medium-low overlap	0.40	0.96	1.00	0.79	1.00	1.00	1.00	1.00	1.00
	strong separation	0.22	0.87	0.97	0.56	1.00	1.00	0.96	1.00	1.00

Table B9: S2, no regimes: share of the replications that select more than one regime, with its standard error, by number of neighbors m, length of the series T and number of locations d.

m	T	d	M	$\Pr(\hat{K} > 1)$
3	120	100	199	0.00 (0.00)
5	60	40	200	0.00 (0.00)
5	60	100	200	0.00 (0.00)
5	60	200	200	0.00 (0.00)
5	120	40	200	0.00 (0.00)
5	120	100	200	0.00 (0.00)
5	120	200	200	0.00 (0.00)
5	120	400	200	0.00 (0.00)
5	365	40	200	0.00 (0.00)
5	365	100	200	0.00 (0.00)
5	365	200	200	0.00 (0.00)
10	120	100	200	0.00 (0.00)

B6.3 Recovery of the partition

Figure B7 and Table B10 give the adjusted Rand index of the partition at the true number of regimes. In the well-separated scenario-variants the partition is recovered almost exactly from $d = 100$ and $T = 120$ on (mean ARI at least 0.99), and with a few poor replications at $d = 40$ and $T = 60$. In S1w-`shr` and S3beta-`shr` the mean ARI ranges from 0.34 to 0.95, grows with d and above all with T, and varies widely from one replication to another. Unlike the selection of K, the recovery of the partition at K_0 hardly depends on the geometry.

Table B10: Mean Adjusted Rand index of the partition at the true number of regimes in the core of the design, by scenario-variant, geometry, T and d.

		$T = 60$			$T = 120$			$T = 365$		
scenario-variant	geometry	$d = 40$	$d = 100$	$d = 200$	$d = 40$	$d = 100$	$d = 200$	$d = 40$	$d = 100$	$d = 200$
`S1s-ind`	total overlap	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	medium-high overlap	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	medium-low overlap	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	strong separation	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
`S1s-shr`	total overlap	0.90	0.96	0.98	0.99	1.00	1.00	1.00	1.00	1.00
	medium-high overlap	0.89	0.96	0.98	0.99	1.00	1.00	1.00	1.00	1.00
	medium-low overlap	0.88	0.96	0.99	0.99	1.00	1.00	1.00	1.00	1.00
	strong separation	0.87	0.97	0.98	1.00	1.00	1.00	1.00	1.00	1.00
`S1w-shr`	total overlap	0.36	0.41	0.47	0.49	0.55	0.64	0.74	0.77	0.89
	medium-high overlap	0.36	0.42	0.48	0.47	0.59	0.67	0.74	0.83	0.95
	medium-low overlap	0.34	0.40	0.45	0.46	0.54	0.63	0.70	0.79	0.94
	strong separation	0.36	0.43	0.46	0.47	0.55	0.62	0.71	0.84	0.93
`S3beta-shr`	total overlap	0.47	0.52	0.58	0.57	0.65	0.73	0.71	0.83	0.85
	medium-high overlap	0.46	0.53	0.60	0.55	0.66	0.76	0.68	0.80	0.85
	medium-low overlap	0.44	0.53	0.57	0.54	0.62	0.74	0.66	0.77	0.83
	strong separation	0.46	0.53	0.59	0.54	0.62	0.77	0.65	0.77	0.94
`S3G-shr`	total overlap	0.86	0.92	0.95	0.96	0.99	1.00	1.00	1.00	1.00
	medium-high overlap	0.83	0.94	0.96	0.96	1.00	1.00	1.00	1.00	1.00
	medium-low overlap	0.83	0.94	0.96	0.96	0.99	1.00	1.00	1.00	1.00
	strong separation	0.83	0.94	0.96	0.96	0.99	1.00	1.00	1.00	1.00
`S3Seta-shr`	total overlap	0.87	0.92	0.94	0.98	0.99	1.00	1.00	1.00	1.00
	medium-high overlap	0.87	0.95	0.97	0.99	1.00	1.00	1.00	1.00	1.00
	medium-low overlap	0.86	0.95	0.97	0.99	1.00	1.00	1.00	1.00	1.00
	strong separation	0.87	0.96	0.97	0.98	1.00	1.00	1.00	1.00	1.00
`S3theta-shr`	total overlap	0.85	0.95	0.97	0.98	1.00	1.00	1.00	1.00	1.00
	medium-high overlap	0.88	0.96	0.98	0.99	1.00	1.00	1.00	1.00	1.00
	medium-low overlap	0.88	0.96	0.98	0.99	1.00	1.00	1.00	1.00	1.00
	strong separation	0.88	0.96	0.98	0.99	1.00	1.00	1.00	1.00	1.00
`S3error-shr`	total overlap	0.87	0.97	0.99	0.97	1.00	1.00	1.00	1.00	1.00
	medium-high overlap	0.86	0.97	0.99	0.97	1.00	1.00	1.00	1.00	1.00
	medium-low overlap	0.86	0.97	0.99	0.98	1.00	1.00	1.00	1.00	1.00
	strong separation	0.85	0.96	0.98	0.98	1.00	1.00	1.00	1.00	1.00

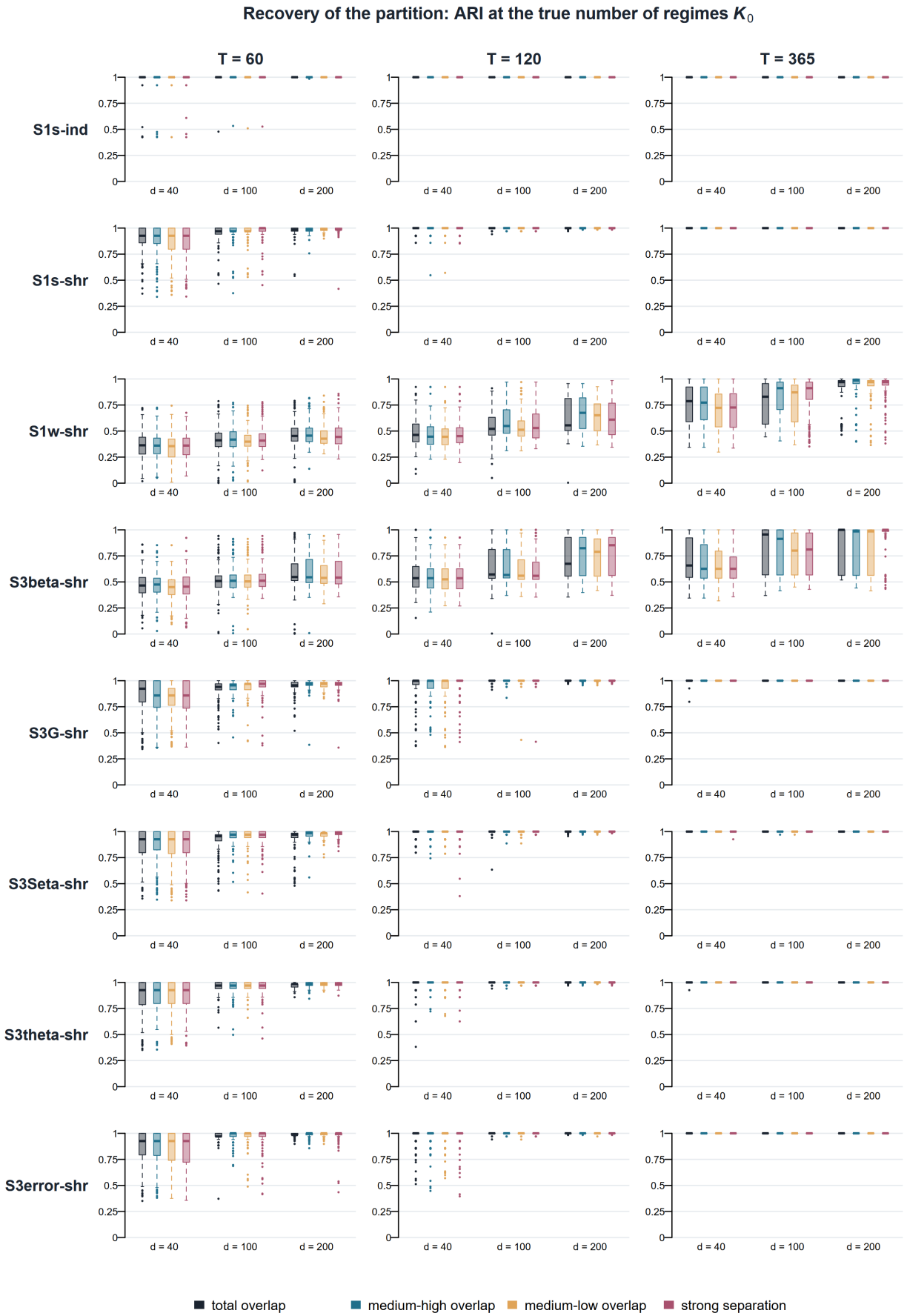


Figure B7: Recovery of the partition in the core of the design: Adjusted Rand index at K_0, one boxplot per cell over its replications, by scenario-variant (rows), T (columns), d and geometry.

B6.4 The parameters

The parameters are recorded for the fit at the true number of regimes, the estimated regimes aligned on the true ones (Section B5). Table B11 gives the reference case, S1s-`ind` at the reference cell, regime by regime, and Table B12 the root mean squared error relative to the true value of every scenario-variant at the reference cell. Figure B8 pools every cell with three regimes and splits the replications by whether the two-step rule selected K_0 in the same replication.

- When the rule selects K_0, the estimates are centered on the truth in every scenario-variant: the median relative error is within 0.4% for the regression coefficients and within 4% for the other parameters. The regression coefficients are the most precise, the persistence, the innovation variance and the range the least.
- When the rule selects the wrong K, the regression coefficients stay centered but are up to four times more dispersed, and the variance components are distorted: the innovation variance and the range are underestimated (median relative error between -5 and -15% and between -4 and -22% across the `shr` variants), and the nugget and the partial sill mostly overestimated (up to $+7$% and $+10$%). These are the replications in which the partition at K_0 is poor — its mean ARI is 0.51 against 0.83 in S1w-`shr`, and 0.68 to 0.88 against at least 0.98 in the other well-separated scenario-variants — and the errors of the partition pass into the parameters. S1s-`ind` selects the wrong K in 19 replications only, with a mean ARI of 0.56.

The figure of the fitted signal in Section 4 of the manuscript scores the selected model, whatever its K, by its fitted signal. When K is right the selected fit and the fit at K_0 nearly coincide, and their root mean squared error is 25 to 42% of that of the pooled fit in the well-separated scenario-variants. When the rule selects too few regimes the error rises to 61 to 64% of the pooled one, still well below it. In S1w-`shr` and S3beta-`shr` the replications in which the rule merges two regimes are those in which the fit at K_0 is poor, and there the merged model fits the signal better than the fit at K_0: median ratio 0.88 against 0.96 in S1w-`shr`, 0.72 against 0.79 in S3beta-`shr`.

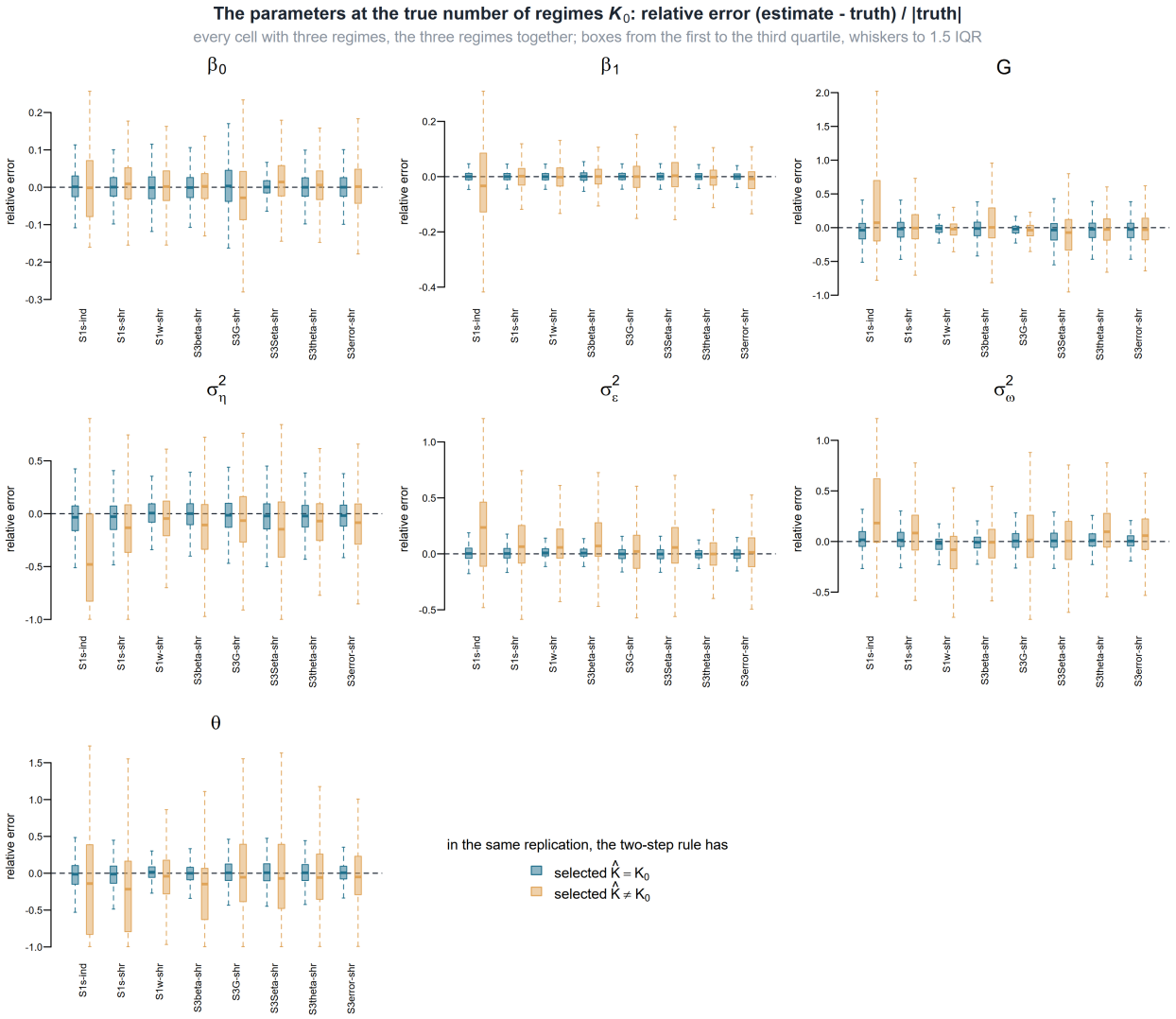


Figure B8: Relative error of the parameters at K_0, (estimate − truth)/|truth|, the three regimes together, over every cell with three regimes, by scenario-variant; in blue the replications in which the two-step rule selected K_0, in orange those in which it did not.

Table B11: The parameters of S1s-`ind` at the reference cell, at the true number of regimes: true value, mean of the estimates, bias and root mean squared error over the replications, regime by regime. β_0 and β_1: intercept and slope; σ^2_η: innovation variance of the latent process.

parameter	regime	M	truth	mean	bias	RMSE
β_0	1	200	2.000	2.010	0.010	0.145
β_0	2	200	2.300	2.310	0.010	0.101
β_0	3	200	2.600	2.603	0.003	0.081
β_1	1	200	0.550	0.550	-0.000	0.014
β_1	2	200	0.850	0.850	0.000	0.016
β_1	3	200	1.150	1.152	0.002	0.017
G	1	200	0.800	0.756	-0.044	0.080
G	2	200	0.500	0.474	-0.026	0.094
G	3	200	0.200	0.186	-0.014	0.100
σ^2_η	1	200	0.108	0.109	0.001	0.022
σ^2_η	2	200	0.300	0.291	-0.009	0.044
σ^2_η	3	200	0.480	0.453	-0.027	0.081
σ^2_ϵ	1	200	0.200	0.199	-0.001	0.013
σ^2_ϵ	2	200	0.350	0.346	-0.004	0.021
σ^2_ϵ	3	200	0.400	0.405	0.005	0.055
σ^2_ω	1	200	0.200	0.202	0.002	0.015
σ^2_ω	2	200	0.150	0.155	0.005	0.021
σ^2_ω	3	200	0.200	0.208	0.008	0.035
θ	1	200	1.500	1.519	0.019	0.234
θ	2	200	2.000	2.063	0.063	0.441
θ	3	200	3.000	2.836	-0.164	1.021

Table B12: Root mean squared error of the parameters at the true number of regimes, relative to the true value and averaged over the three regimes, for every scenario-variant at the reference cell.

scenario-variant	β_0	β_1	G	σ^2_η	σ^2_ϵ	σ^2_ω	θ
`S1s-ind`	0.05	0.02	0.26	0.17	0.09	0.13	0.24
`S1s-shr`	0.05	0.02	0.24	0.17	0.08	0.13	0.23
`S1w-shr`	0.06	0.05	0.12	0.25	0.21	0.24	1.34
`S3beta-shr`	0.05	0.05	0.39	0.30	0.22	0.29	0.51
`S3G-shr`	0.07	0.02	0.09	0.19	0.08	0.13	0.22
`S3Seta-shr`	0.04	0.02	0.27	0.19	0.08	0.13	0.21
`S3theta-shr`	0.05	0.02	0.24	0.17	0.07	0.11	0.22
`S3error-shr`	0.05	0.02	0.23	0.17	0.07	0.08	0.16

B6.5 Robustness

Figure B9 and Tables B13–B14 change one factor of the reference cell at a time. The number of neighbors of the Potts graph changes nothing, as expected under the automatic scale of the penalty. Unbalanced regimes leave the well-separated scenario-variants at or above 96% and hurt the weak ones, S1w-`shr` falling from 25% to 10%, since the smallest regime carries half the evidence of a balanced one. A network of $d = 400$ locations raises the weak scenario-variants to 86% and 76% and declares no false regimes in S2, but the weak scenario-variants select four regimes in 12% and 19% of the replications: the over-selection on wide networks discussed in Section B5 does not appear where there are no regimes, and appears where the regimes are weakly separated and overlap.

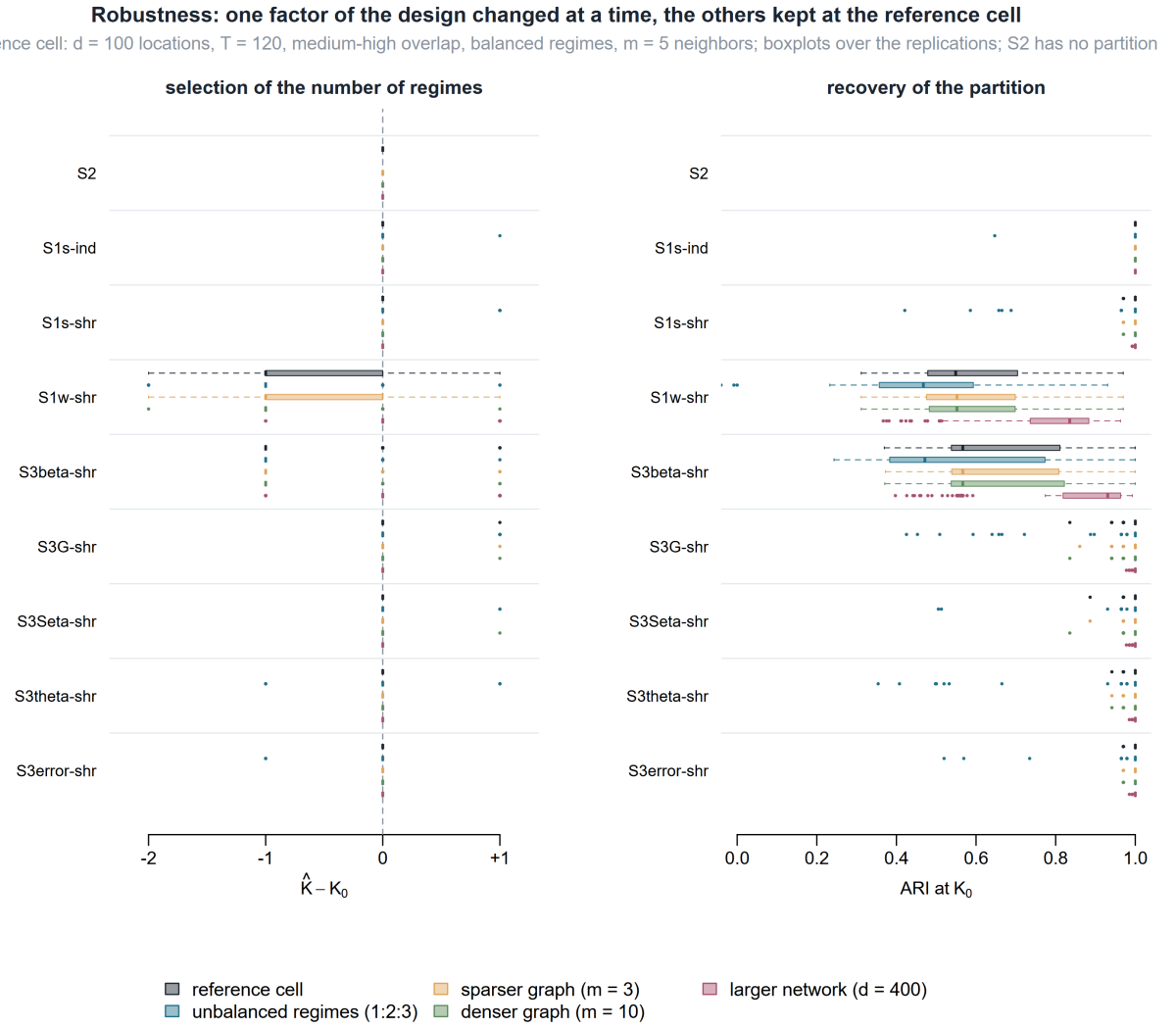


Figure B9: Robustness at the reference cell ($d = 100$, $T = 120$, medium-high overlap, balanced sizes, $m = 5$): boxplots over the replications of the selection error $\hat{K} - K_0$ (left) and of the ARI at K_0 (right), when one factor of the reference cell is changed and the others are kept.

Table B13: Robustness at the reference cell: share of the replications selecting the true number of regimes.

scenario-variant	reference	unbalanced	knn = 3	knn = 10	n = 400
S2	1.00	—	1.00	1.00	1.00
S1s-ind	1.00	0.99	1.00	1.00	1.00
S1s-shr	1.00	0.97	1.00	1.00	1.00
S1w-shr	0.25	0.10	0.25	0.23	0.86
S3beta-shr	0.22	0.23	0.22	0.20	0.76
S3G-shr	0.99	0.96	0.99	0.99	1.00
S3Seta-shr	1.00	0.99	1.00	0.99	1.00
S3theta-shr	1.00	0.97	1.00	1.00	1.00
S3error-shr	1.00	0.98	1.00	1.00	1.00

Table B14: Robustness at the reference cell: mean ARI at the true number of regimes.

scenario-variant	reference	unbalanced	knn = 3	knn = 10	n = 400
S2	—	—	—	—	—
S1s-ind	1.00	1.00	1.00	1.00	1.00
S1s-shr	1.00	0.99	1.00	1.00	1.00
S1w-shr	0.59	0.49	0.59	0.59	0.78
S3beta-shr	0.66	0.56	0.66	0.66	0.85
S3G-shr	1.00	0.98	1.00	1.00	1.00
S3Seta-shr	1.00	0.99	1.00	1.00	1.00
S3theta-shr	1.00	0.98	1.00	1.00	1.00
S3error-shr	1.00	0.99	1.00	1.00	1.00

B6.6 The spatial penalty

Figure B10 gives the ARI at K_0 against the penalty, at the reference cell. Up to $\phi = 0.2$ the penalty leaves the partition of the well-separated scenario-variants unchanged and improves slightly that of the weak ones (from 0.56 to 0.61 in S1w-**shr** and from 0.62 to 0.67 in S3beta-**shr**); beyond, it oversmooths the borders and the ARI falls, to about 0.5 at $\phi = 1$. When the rule selects K_0, it selects $\phi = 0$ in 79 to 100% of the replications of the well-separated scenario-variants, while in the weak ones the selected penalty spreads over the grid, with $\phi = 0.2$ the most frequent in S1w-**shr**.

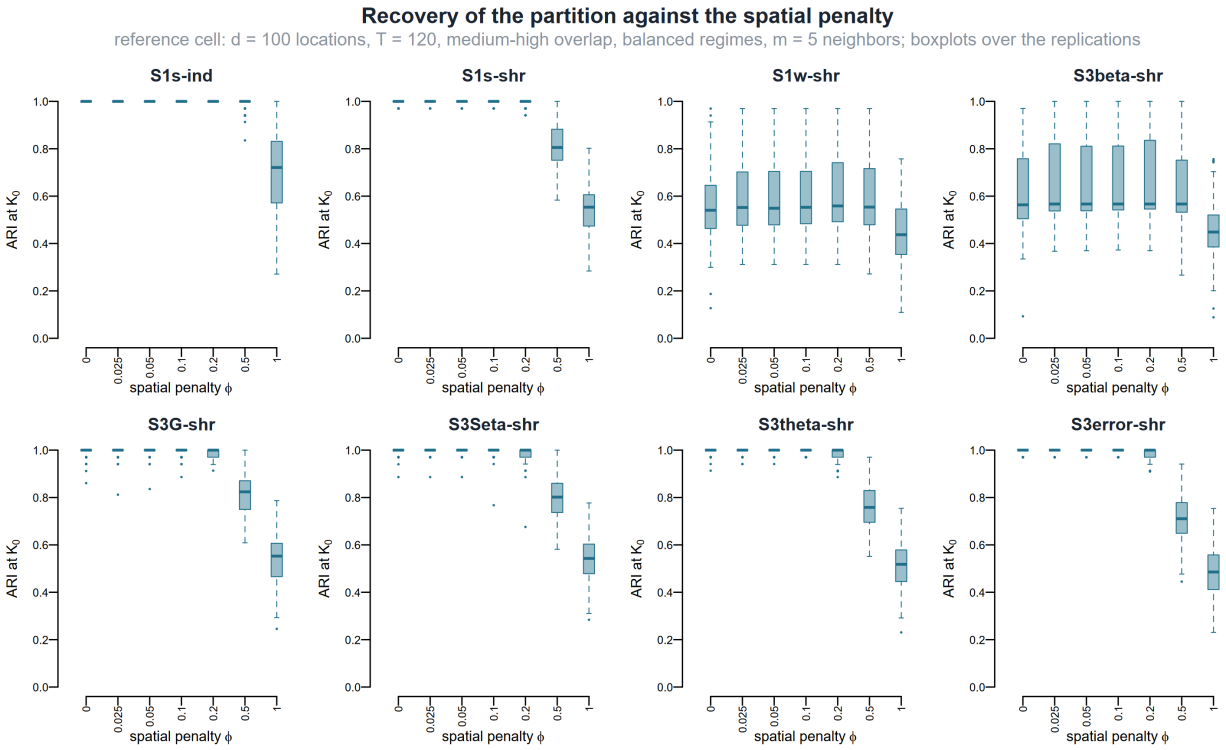


Figure B10: ARI at K_0 against the spatial penalty ϕ: boxplots over the replications of the reference cell ($d = 100$, $T = 120$, medium-high overlap, balanced sizes, $m = 5$), one panel per scenario-variant.

B6.7 The information criteria in the two-step rule

Table B15 gives the share of the replications that select K_0 with each of seven criteria. The AIC, the AICc and the GCV behave alike and select K_0 more often than the BIC in the scenario-variants that the BIC under-selects (0.32–0.33 against 0.26–0.28 in S1w-`shr` and S3beta-`shr`, 0.96–0.97 against 0.89–0.94 in S1s-`shr`, S3G-`shr`, S3theta-`shr` and S3error-`shr`), at the price of more over-selection (4–7% against 2–4% in the weak scenario-variants, 4% and 3% at $d = 400$). The HQ and the KIC lie between the AIC and the BIC, and the EBIC under-selects most. No criterion ever splits S2.

Table B15: Share of the replications selecting the true number of regimes when the two-step rule is run with each criterion, over every cell, by scenario-variant.

scenario-variant	AIC	AICc	GCV	HQ	KIC	BIC	EBIC
S2	1.00	1.00	1.00	1.00	1.00	1.00	1.00
S1s-ind	1.00	1.00	1.00	1.00	1.00	1.00	1.00
S1s-shr	0.97	0.97	0.97	0.96	0.97	0.93	0.89
S1w-shr	0.33	0.33	0.33	0.31	0.32	0.28	0.21
S3beta-shr	0.32	0.32	0.32	0.30	0.31	0.26	0.19
S3G-shr	0.97	0.97	0.97	0.96	0.97	0.94	0.89
S3Seta-shr	0.98	0.98	0.98	0.98	0.98	0.97	0.95
S3theta-shr	0.96	0.96	0.96	0.94	0.95	0.89	0.80
S3error-shr	0.96	0.96	0.96	0.95	0.96	0.92	0.86

B7 Reproduction code

The study is reproduced by three scripts in the folder `papers/sc-stem-local-leader/code` of the repository of `Stem` (Cameletti et al., 2026), https://github.com/PaoloMaranzano/Stem, which also holds the results of the 200 replications. The scripts use `Stem` as an ordinary dependency: they check that the installed version is the one the results were computed with, and install it otherwise. Every routine that uses randomness takes an explicit seed and restores the state of the random number generator on exit, so the study reproduces exactly.

`Sims_00_Setup.R` implements the generator and the replication of the procedure, and defines the scenarios, the parameter values and the blocks described here; the number of locations is called `n` there.

`Sims_01_Run.R` runs the replications under the time limit and writes their results.

`Sims_02_Analysis.R` turns the results into the tables and figures of Section 4 of the manuscript and of Section B6, and produces Figures B1–B5 and Tables B1, B5 and B6 from the definitions of `Sims_00_Setup.R`, so that the description of the design cannot disagree with the design that is run.

References

Michela Cameletti, Francesco Caccia, and Paolo Maranzano. *Stem: Spatio-Temporal Expectation-Maximization Models and their Spatially-Clustered Extension*, 2026. URL https://github.com/PaoloMaranzano/Stem. R package version 2.0.0.

Tim P. Morris, Ian R. White, and Michael J. Crowther. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11):2074–2102, 2019. doi: 10.1002/sim.8086.